- Each task has its own model
- Closed environments with simple scenarios
- Closed-set vocabulary (fixed categories)
- No shared knowledge across tasks

Previous work

Task-specific Learning

Video Segmentation Foundation Model

VIS

Model/Data/Loss #1

VOS

Model/Data/Loss #2

PVOS

Model/Data/Loss #3

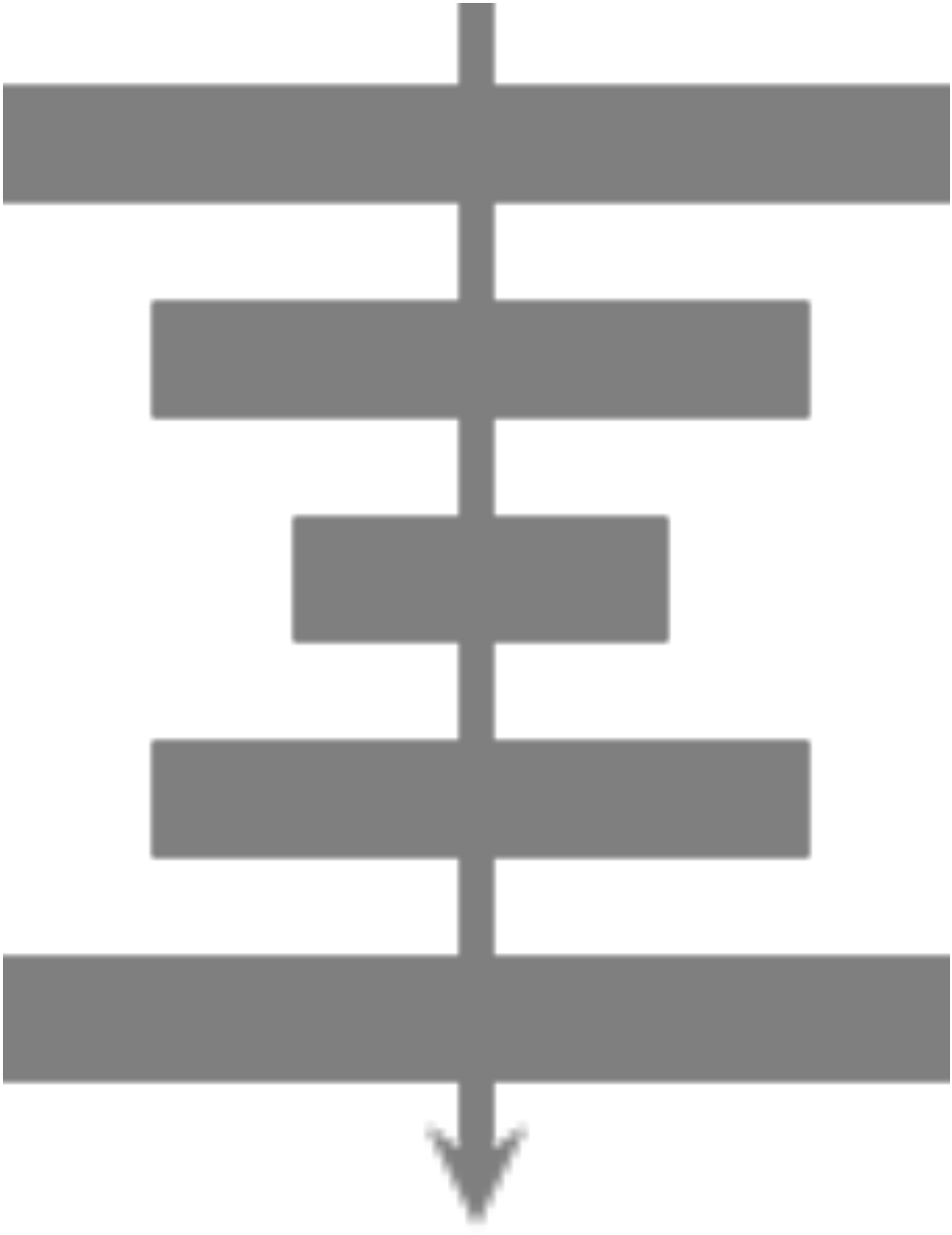
RefVOS

Model/Data/Loss #4

Poor generalization

User #1:

Segment all objects



Category names:

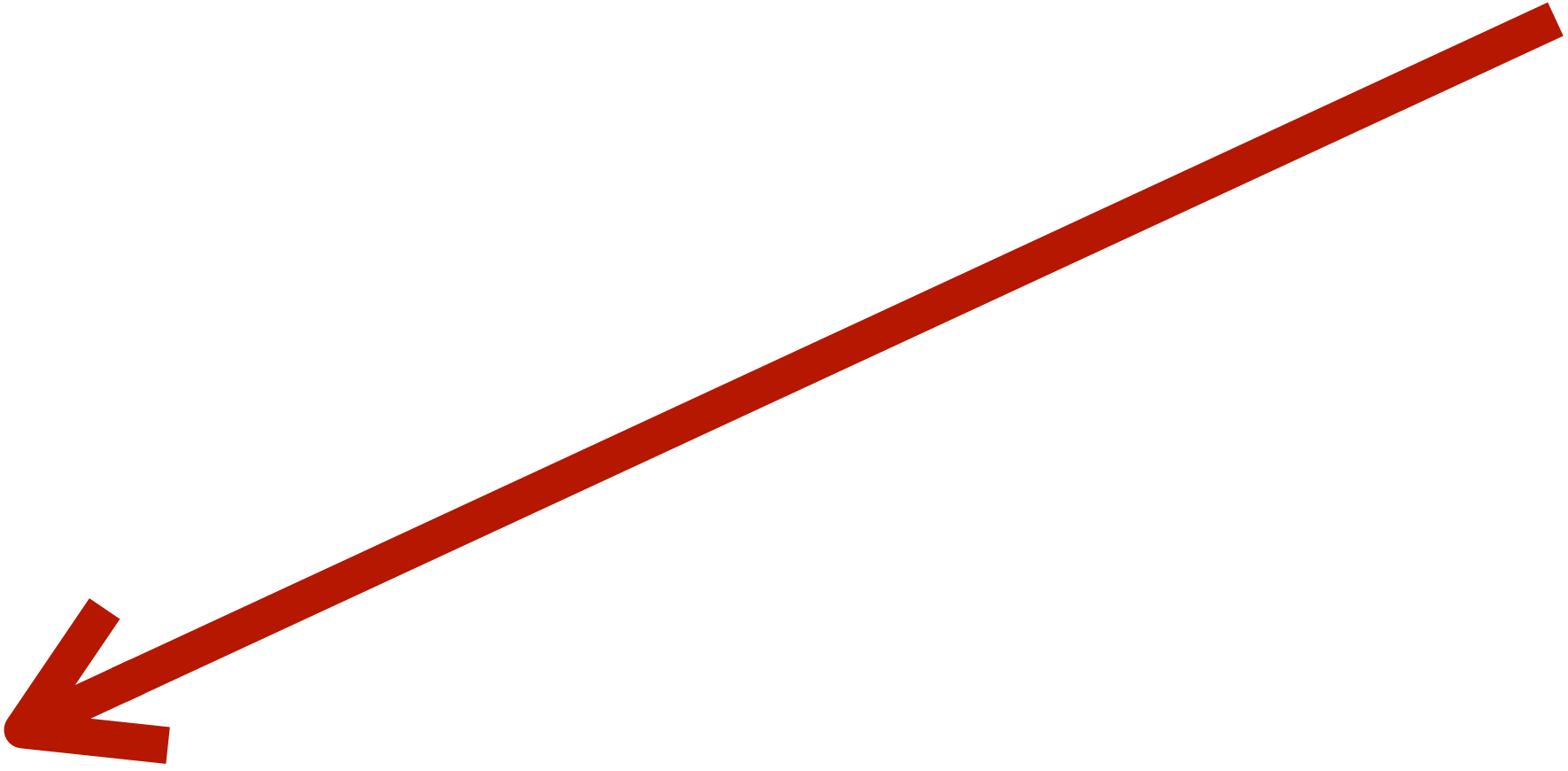
Person, Roof, Dog, Washer, ...

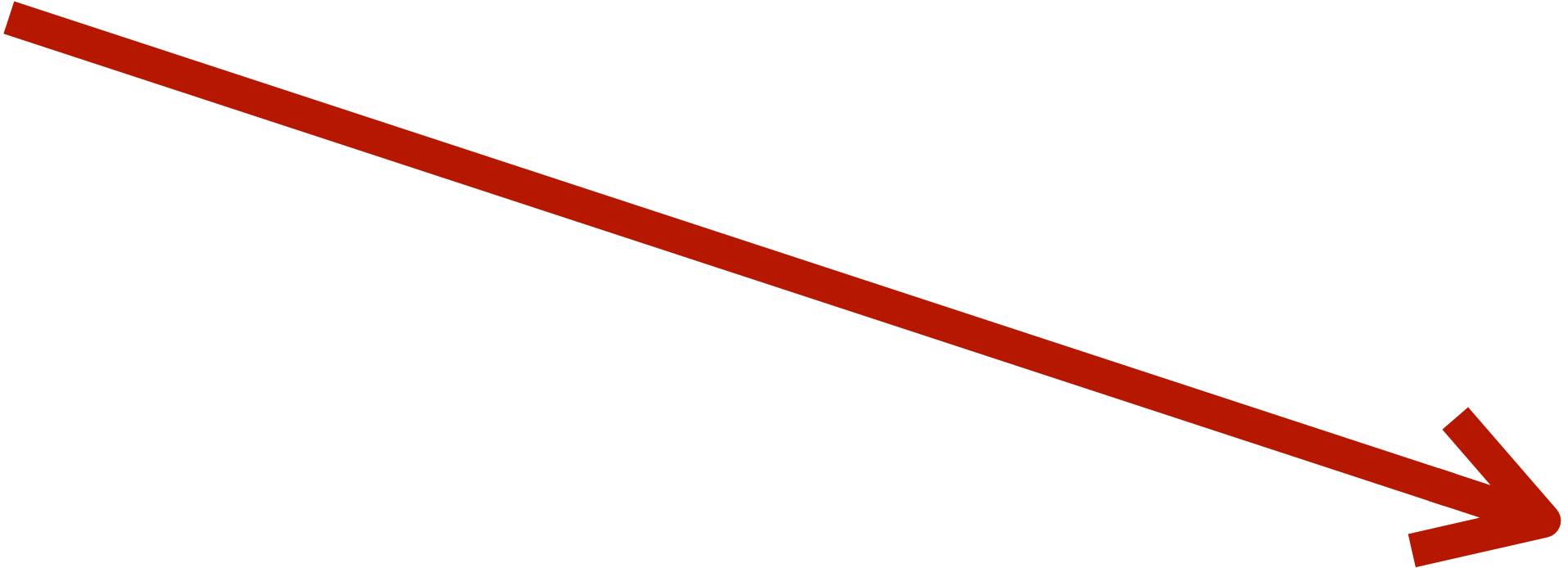
Text prompt:

The person closest to the camera

Training

Inference





Does not work!!!

DIGITAL

- Open-word vocabulary

- Scalability to long videos

- Open environments with complex scenes

- One model can handle all tasks

Cross Task Learning

Model/Data/Loss #1

VIS

Model/Data/Loss #2

VOS

PVOS

Model/Data/Loss #3

RefVOS

Model/Data/Loss #4

VOS, RefVOS, PVOS, VIS, VSS, VPS

One model / All data / All losses

Foundation Model

across prompt types and time

Consistent understanding

Segment all objects

User #1:

Person, Roof, Dog,

Category names:

Washer, ...

The person closest

Text prompt:

to the camera

Inference

Does not

work!!!

Poor generalization